SAY WATT ROBOTICS · FIELD LOG

Week 2.6 — Does It Fit? — End Cap Seals and the Whole Profiler in CAD

Sunday, October 25, 2026 · 3:00 – 5:00 PM · 5 students

Five of us at 2.6 — Trenton, Isaac, Rebecca, Adrian, and Frankie — from 3:00 to 5:00. A small table and a focused one. Mechanical worked on the fit of the end cap seals in CAD, picking up where 2.5 left off: last week the team calculated the depth of the O-ring groove, and this week they worked out how the cap, the ring, and the tube actually come together. They also started putting the whole profiler together as one CAD assembly — hull, end caps, and buoyancy engine in the same model — which is where parts that look perfect on their own find out whether they collide with their neighbors. Electrical kept moving in KiCad on the circuit board that will run the float.

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STUDENTS AT THE SESSION	FULL PROFILER ASSEMBLY STARTED IN CAD	TRACKS MOVING — MECHANICAL CAD AND KICAD

WHAT WE LEARNED

- Model the assembly, not just the parts. A part can look perfect alone and still collide with its neighbor; the full assembly is where interference shows up before anything gets printed.
- A seal is a fit, not a part. An O-ring only works when the groove, the ring, and the surface it presses against all line up, and CAD is the cheapest place to check that.
- A small table still moves the work. Five people kept both the mechanical model and the circuit board going.

DECISIONS MADE

- The profiler is being modeled as a full assembly, not as a pile of separate parts.
- End cap seal fitment gets settled in CAD before the next print.

STUDENT LEARNING OUTCOMES

- Worked out end cap seal fitment in CAD.
- Started a full CAD assembly of the profiler.
- Continued the profiler's circuit board in KiCad.

LOOKING AHEAD

Bring the buoyancy engine into the full assembly, keep the KiCad board moving, and get the end cap ready to print.

MADE POSSIBLE BY